Qiskit Fall Fest 2026

Challenge 3: Can you change the correlation?

Problem Statement - Two qubits can be prepared in a state where their measurement outcomes are strongly correlated. Your task is to modify a two-qubit circuit and investigate whether you can change the relationship between the two measurement outcomes.

Your Task - Using Qiskit:

1. Create a two-qubit quantum circuit.
2. Start from the Bell-state circuit demonstrated during the workshop.
3. Add or modify one quantum gate.
4. Measure both qubits.
5. Execute the circuit for 100 shots.
6. Observe the measurement results.
7. Display the results using a histogram.
8. Compare the results with the original circuit.
9. Explain how the modification changed the measurement outcomes.

Think Before You Run - Before executing your modified circuit, predict: Which outcomes do you expect? Will the two qubits still produce the same outcomes? Will 00 and 11 dominate? Could 01 and 10 become more frequent?

Design Constraints - Your circuit must:

- Use exactly 2 qubits.
- Measure both qubits.
- Start from the Bell-state circuit.
- Add or modify only one quantum gate.
- Use no more than 4 quantum gates before measurement.
- Be created and executed using Qiskit.
- Use 100 shots.

Submission - Your notebook must include:

- Original circuit.
- Modified circuit.
- Prediction before execution.
- Measurement results for both circuits.
- Histograms.
- A short explanation of the observed change.